

Generation Interconnection System Impact Study Report

For Queue Project Q21-1873

Oakwind 100 kV
October 17, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1873. The Interconnection Customer proposes a 122.0 MW battery energy storage facility interconnecting at the Oakwind 100 kV substation in the Ohio area. The request was received on August 31, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 632 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 21 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$34,573,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Marcos - Cleveland Thermal 100.0 kV line	Scranton - Marcos 100.0 kV	189.1	130	11.4
Marcos - Scranton 100.0 kV Line	Cleveland Thermal - Marcos 100.0 kV Line	163.7	130	9.5
Lincoln Beach - Hills 100.0 kV Line	Imperial Substation - Chamberlin 230.0 kV	163.5	127	5.7
Macedonia - N Wellsville 100.0 kV line	Rhinebeck - Lincoln Beach 100.0 kV line	129.9	135	10.0
Burney Forest Power - Marcs 100.0 kV L	Galeville - Marcos 100.0 kV Line	125.9	108	6.9
Dickinson - Ave Maria S 100.0 kV	Galeville - Marcos 100.0 kV	124.9	164	10.6
Rhinebeck - Lincoln Beach Substation 100.0 kV	N Wellsville - Macedonia 100.0 kV	124.3	148	10.8
West 41St St - Loring Power Plant 100.0 kV	Marcos - Galeville 100.0 kV L	122.5	112	7.0
Jaris - Oakwind 100.0 kV Line 1	Jarvis - Oakwind Substation 100.0 kV	122.0	148	74.6
Oakwind - Jarvis 100.0 kV	Jarvis - Oakwind 100.0 kV	122.0	148	74.6
Macedonia - South Belridge Cogen 100.0 kV	Lincoln Beach - Rhinebeck 100.0 kV Line	120.1	128	10.0
Juniper - Rhinebeck Substation 100.0 kV L	Macedonia - N Wellsville 100.0 kV	118.8	125	10.8
Chasm - Bulterville 100.0 kV	Hylebos 500/230kV Auto	113.7	116	15.2
Juniper - South Belridge Cogen 100.0 kV Line	Rhinebeck - Lincoln Beach 100.0 kV	110.8	119	10.0
Galeville - Marcos 100.0 kV	Eufaula - Imperial Substation 500.0 kV	109.9	297	10.2

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Ave Maria S - Imperial 100.0 kV	Ave Maria outh - Imperial 100.0 kV Cir. 2	109.8	149	6.8
Ave Maria S - Imperial 100.0 kV Ckt 2	Ave Maria S - Imperial 100.0 kV	109.8	149	6.8
Morristown Substation - High Bridge Substation 100.0 kV	Morristown - High Bridge Substation 100.0 kV Cir. 2	107.5	145	5.7
Morristown - High Bridge 100.0 kV Line 2	High Bridge - Morristown 100.0 kV Circuit 1	107.5	145	5.7
Morristown - Winger 100.0 kV	Morristown - Winger 100.0 kV #2	102.2	181	5.8
Morristown - Winger 100.0 kV	Morristown - Winer 100.0 kV #1	102.2	181	5.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cleveland Thermal - Marcos Substation 100.0 kV	Rebuild line with higher-capacity conductor	\$1,177,000	51% (\$602,000)
Scranton - Marcos 100.0 kV Line	Rebuild line with higher-capacity conductor	\$1,845,000	80% (\$1,467,000)
Lincoln Beach - Hills 100.0 kV Line	Rebuild line with higher-capacity conductor	\$2,630,000	100% (\$2,630,000)
Macedonia - N Wellsville 100.0 kV line	Rebuild line with higher-capacity conductor	\$1,903,000	100% (\$1,903,000)
Burney Foest Power - Marcos 100.0 kV	Rebuild line with higher-capacity conductor	\$2,414,000	100% (\$2,414,000)
Dickinson - Ave Maria S 100.0 kV	Rebuild line with higher-capacity conductor	\$2,689,000	100% (\$2,689,000)
Rhinebeck - Lincoln Beach Substation 100.0 kV	Rebuild line with higher-capacity conductor	\$2,687,000	100% (\$2,687,000)
W 41St St - LoringPower Plant 100.0 kV	Rebuild line with higher-capacity conductor	\$5,917,000	100% (\$5,917,000)
Jaris - Oakwind 100.0 kV Line 1	Rebuild line with higher-capacity conductor	\$1,192,000	100% (\$1,192,000)
Jarvis - Oakwind 100.0 kV Line 2	Rebuild line with higher-capacity conductor	\$1,524,000	100% (\$1,524,000)
Macedonia - South Belridge Cogen 100.0 kV	Rebuild line with higher-capacity conductor	\$1,703,000	100% (\$1,703,000)
Juniper - Rhinebeck Substation 100.0 kV L	Rebuild line with higher-capacity conductor	\$1,641,000	100% (\$1,641,000)
Chasm - Bulterville 100.0 kV	Rebuild line with higher-capacity conductor	\$2,129,000	57% (\$1,205,000)
Juniper - S Belridge Cogen 100.0 kV	Rebuild line with higher-capacity conductor	\$1,753,000	100% (\$1,753,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Galeville - Marcos 100.0 kV	Reconductor line	\$1,393,000	81% (\$1,133,000)
Ave Maria S - Imperial 100.0 kV Line 1	Reconductor line	\$1,410,000	100% (\$1,410,000)
Ave Maria S - Imperial 100.0 kV Ckt 2	Reconductor line	\$1,609,000	100% (\$1,609,000)
Morristown - High Bridge 100.0 kV Cir. 1	Reconductor line	\$261,000	100% (\$261,000)
Morristown - High Bridge 100.0 kV Cir. 2	Reconductor line	\$359,000	100% (\$359,000)
Morristown - Wingr 100.0 kV	Reconductor line	\$272,000	100% (\$272,000)
Morristown - Winger 100.0 kV	Reconductor line	\$202,000	100% (\$202,000)
	Total allocated to project		\$34,573,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.